

What Is Being Trained On What

The models, the eight datasets, the six splits, and the ladder — and exactly how they fit together. Written for a reader with no background in chemistry, computing or statistics.

1. Start here: the one thing that surprises everyone

There is **not one model** in this project. There are **thousands**.

A common guess is that we built one clever program, showed it eight datasets, and it learned all of them. That is not what happens. Each dataset asks a completely different question with a different kind of answer, so **each dataset gets its own separately trained model**. And because we test each design on six different ways of splitting the data, each design is trained **six times per dataset**.

One design $\times$ 8 datasets $\times$ 6 splits = **48 trained models**, just to evaluate *one* idea once. Across the whole project we have trained and archived **1,606 models** covering **62 different designs**.

The rest of this document explains each of those three multipliers: the datasets, the splits, and the designs.

2. First, some words you need

Word	What it means, in one line
Molecule	A substance — a medicine, a poison, a sugar. Made of atoms joined by bonds.
SMILES	A way of writing a molecule as text, so a computer can read it. Aspirin is <chem>CC(=O)Oc1ccccc1C(=O)O</chem> .
Property	Something true about a molecule that we would like to know without doing the experiment — is it toxic, does it dissolve.
Dataset	A list of real molecules with a property that scientists actually measured in a lab.
Task	One single question inside a dataset. Some datasets ask one question; one asks 27.
Model	A program that reads a molecule and guesses a property. It learns by looking at the measured examples.
Training	Showing the model molecules whose answers we know, and repeatedly nudging it to be less wrong.
Split	Dividing a dataset into a part to learn from and a part to be tested on, so the test is honest.

3. The eight datasets — eight different questions

These are public collections that chemists have used as a shared yardstick for years. We did not create them. Each one contains real molecules with real measured answers.

Dataset	The question it asks	Answer looks like	Questions	Molecules
tox21	Is this molecule toxic? (12 separate biological alarms)	yes / no, 12 times over	12	7,823
bbbp	Can it get from the blood into the brain?	yes / no	1	2,039
clintox	Did it pass or fail human clinical trials?	yes / no, twice	2	1,480

Dataset	The question it asks	Answer looks like	Questions	Molecules
esol	How well does it dissolve in water?	a number (log mol/L)	1	1,128
lipophilicity	Does it prefer fat or water?	a number (logD)	1	4,200
bace	Does it block an enzyme linked to Alzheimer's?	yes / no	1	1,513
sider	What side effects does it cause? (27 body systems)	yes / no, 27 times over	27	1,427
freesolv	How much energy does it take to dissolve it in water?	a number (kcal/mol)	1	642

Two things to notice.

- **The answers are different kinds of thing.** Five datasets want a yes/no ("classification"). Three want a number ("regression"). A model that outputs yes/no cannot output a number, so these cannot share one model even in principle.
- **They are small.** The biggest has about 7,800 molecules; the smallest has 642. In modern machine learning that is tiny. This matters more than anything else in this project: on data this small, results wobble a lot, and telling a real improvement from luck is the hard part.

4. How a model sees a molecule: the three “views”

A computer cannot read a drawing of a molecule. It needs numbers. There are several reasonable ways to turn a molecule into numbers, and none of them is obviously best — so this project uses three at once and calls them **views**.

View	What it does	Like...
Graph	Treats the molecule as a network of atoms joined by bonds, and lets information flow between neighbouring atoms.	Reading a map of a city by following its roads.
Sequence	Feeds the SMILES text into ChemBERTa — a language model that was pre-trained on millions of molecule strings, the same idea as ChatGPT but for chemistry notation.	Reading the molecule as a sentence.
Descriptors	Computes ~217 standard chemical measurements (weight, greasiness, number of rings...) plus a 1,024-slot checklist of which small fragments are present.	Filling in a fact sheet about the molecule.

Why three? Each view is blind to something. The fact sheet does not know how the atoms are connected. The map does not know the molecule's overall weight. The hope behind this project is that **combining** them beats any one of them. Testing that hope honestly is what the whole project is about.

5. The splits: why six, and what “scaffold” means

If you train a model and test it on the same molecules, it will look brilliant and be useless — it has memorised the answers. So the data is divided up first.

Part	Share	What it is for
Training set	80%	The model learns from these. It sees both the molecules and the answers.

Part	Share	What it is for
Validation set	10%	Used <i>during</i> training to decide when to stop and to make every choice. The model never learns from these.
Test set	10%	Touched once, at the very end, to report the score. Never used to make any decision.

What a scaffold split is

A **scaffold** is a molecule's skeleton — the core ring structure left when you strip off the decorations. Thousands of real medicines share a few hundred scaffolds; they come in families.

If you split the data **randomly**, close cousins from the same family land in both the training set and the test set. The model then gets tested on molecules that are nearly identical to ones it studied. It scores brilliantly and has learned nothing useful.

A **scaffold split** keeps every member of a family together. If one cousin is in the training set, all of them are. The test set is made of skeletons the model has **never seen in any form**. It is a much harder test — and it is the honest one, because a real chemist wants predictions about genuinely new molecules.

And why six of them

There is more than one way to divide molecules into families and hand them out. Change the way you do it and the same model, with the same code, can score noticeably differently — by pure luck of the draw.

- **One** is the standard split everybody in the field uses, so our numbers can be compared with published ones.
- **Five** more are generated by us with different random starting points — same rules, different shuffles.

Every model is trained on all six. We then report the **average across the five** and how much they disagreed. A difference that only shows up on one split is noise; one that survives all five is a finding.

This is the single most important design decision in the project.

6. The ladder: five ways to combine the three views

Once you have three views, you have to merge them into one answer. There are many ways. We built five, from the simplest possible to the most elaborate, each adding exactly one idea to the one before. That staircase is what we call **the ladder**.

Rung	What it adds	Extra parts to learn
1. concat	Just stack the three views end to end. Nothing is learned about how to combine them. The floor of the ladder.	none
2. gated	Learns, for each molecule, how much to trust each view — like a volume knob per view.	16,513
3. xattn	Cross-attention: lets each view look at the other two and update itself before merging.	1,054,208
4. bilinear	Multiplies pairs of views together, so the model can react to <i>combinations</i> rather than to each view separately.	99,072
5. proposed	All of it at once: cross-attention, then multiplication, then the volume knobs. Our own design.	1,169,793

Why build a ladder instead of just the best one? Suppose we only built rung 5, and it won. We would not know *why*. Was it the attention? The multiplication? The volume knobs? Or none of them — just the extra parts? Building every rung and testing all five the same way is the only way to find out which idea actually did the work. That is an **ablation**: remove one piece at a time and see what breaks.

7. Putting the three multipliers together

Now the arithmetic that produces thousands of models.

	How many	Why
Designs on the ladder	5	concat, gated, xattn, bilinear, proposed
× Datasets	8	each asks a different question, so each needs its own model
× Splits	6	1 standard + 5 of ours, so we can tell a finding from luck
= Models for one ladder	240	and that is one experiment, run once

And the ladder is only part of it. We also trained the three views on their own (to see whether combining helps at all), the pipeline this project inherited (to measure improvement against), two published methods from other research groups (to check we are not beating only ourselves), and the whole ladder a second time with the language model un-frozen. Each of those is another 48 models.

Total archived so far: **1,606 trained models** across **62 designs**, almost all of them tested on all 8 datasets and all 6 splits. Roughly 30 hours of computing time, mostly on free cloud GPUs.

8. What actually happens during training

Step	In one line
1. Feed a batch	Show the model 128 molecules at once and let it guess their properties.
2. Measure the error	Compare every guess with the measured truth and turn the difference into one number.
3. Nudge	Adjust every internal setting slightly in whichever direction reduces that error.
4. Repeat	Go through the whole training set again. One full pass is called an epoch .
5. Check on the validation set	After each epoch, score molecules it did not learn from.
6. Stop when it stops improving	If 15 epochs pass with no improvement, stop and keep the best version. This is early stopping , and it prevents memorising.
7. Score once on the test set	One number, reported. Nothing is changed after seeing it.

Everything is trained the same way. Same learning speed, same batch size, same stopping rule, for every model in every table — written down in one file (`configs/shared.yaml`) and checked automatically. If we tuned our own design and left the others at default settings, our design would win and the comparison would be worthless. It would be measuring effort, not ideas.

9. Every model in the project, and its job

Name	What it is	Why it exists
gin_ref	The simple graph model this project inherited.	The thing to beat. Every improvement is measured against it.
gine	A bigger graph model that also reads the bonds, not just the atoms.	Tests whether a stronger graph reader helps. 1,070,852 parts vs 207,360.
desc	The chemical fact sheet alone, through a small network.	The unglamorous baseline. Often the one to beat.
seq_frozen	ChemBERTa used as-is, not adjusted.	The cheap way to use a language model.
lora	ChemBERTa with a small trainable add-on.	Tests whether adjusting the language model is worth ~11 hours of GPU time.
fuse_concat ... fuse_proposed	The five rungs of the ladder.	The main experiment: does combining views help, and which combining trick does the work?
attentivefp	A published graph model from another research group.	Checks we are not just beating our own weak baseline.
chemprop	A widely used published tool from another group.	Same reason. It is the one a chemist would actually reach for.
rf / gnn / trf / hybrid / ens	The five models of the inherited pipeline.	The starting point, kept so the before/after is real.

10. So which model does the demo use?

The Streamlit app serves **fuse_proposed** — rung 5, our own design — with one separately trained copy per dataset, all trained on the standard scaffold split. Type a molecule and it runs eight models, one per property.

Alongside every prediction the app shows two things most demos hide: a **confidence range** (or, for yes/no questions, whether the model can decide at all at 90% confidence), and the model's **measured accuracy on the test set**, so a viewer can see how much the number is worth.

11. What we actually found — the honest version

First, a hidden clue: in two datasets (ClinTox and BBBP) the way a molecule is *written* gives the answer away, and the language model was reading it. Once those molecules were rewritten in one standard form, those scores fell sharply.

With the clue removed, our elaborate design (rung 5, over a million extra parts to learn) **did not reliably beat** the simple volume-knob design (rung 2, about sixteen thousand) or the inherited graph model, and no rung of the ladder beat the plain chemical fact sheet. The two published methods from other groups did not reliably beat the inherited model either.

This is a real result, not a failure. The project was built to measure honestly, and what it measured is that our own improvement did not survive being tested on six splits instead of one. We are reporting it. A measuring instrument that only ever confirms the person holding it is not a measuring instrument.

12. Questions a reader usually asks next

Why not train one model on all eight datasets together?

Because they answer different kinds of question in different units — yes/no versus a number in kcal/mol. Sharing knowledge between them is a real research area, but it is a different project, and mixing it in would make it impossible to say which change caused what.

Why six splits and not more?

Each extra split costs another full round of training on every dataset. Six gives us a stable average and a statistical test strong enough to detect the size of difference the field reports, which is what we need. More would be better and would cost more than we have.

Is the test set really never used?

Yes. Every choice — when to stop, which version to keep, which design to report — is made on the validation set. Fixing that was one of the first things this project did: the inherited pipeline had been picking its winner by looking at the test score, which quietly inflates every number it reports.

Why is the ladder trained twice?

Once with the language model frozen (fast) and once with it learning too (about eleven hours of GPU time). Freezing it is a shortcut, and we wanted to know whether the shortcut cost anything. It did not.

How do you know the demo app is using the same features as training?

There is an automatic check (`scripts/check_deploy.py`) that takes molecules from the training data, runs them through the app's own code, and demands the numbers match exactly. A model fed slightly wrong inputs does not crash — it returns confident nonsense — so this is checked rather than assumed.

If you remember one sentence: each of the eight datasets gets its own model, each model is trained six times on six different honest splits, and the five rungs of the ladder exist so that when something works we can say *which part* of it worked — which is how 1,606 trained models turn into one trustworthy answer.